AI CLUSTERS



End-to-End AI backend fabric design

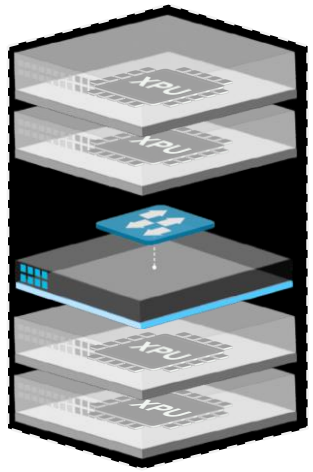
Sandeep Balani *Principal Product Manager, CSG, Broadcom*

Mehak Mahajan *Senior Director of Engineering, CSG, Broadcom*



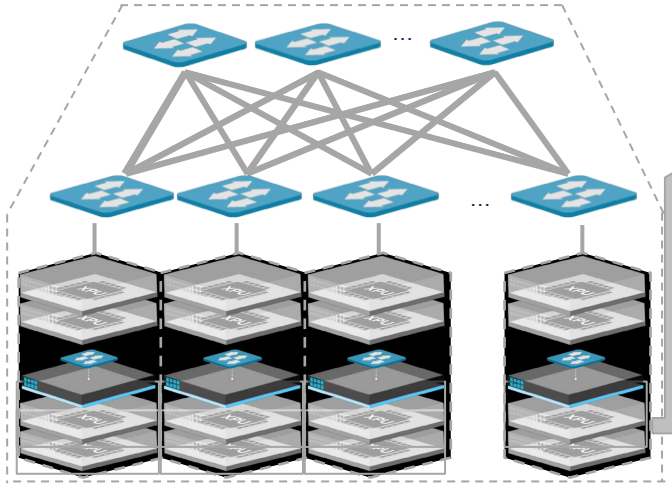
AI Networks – Scale-up, Scale-Out, Scale-Across

In Rack



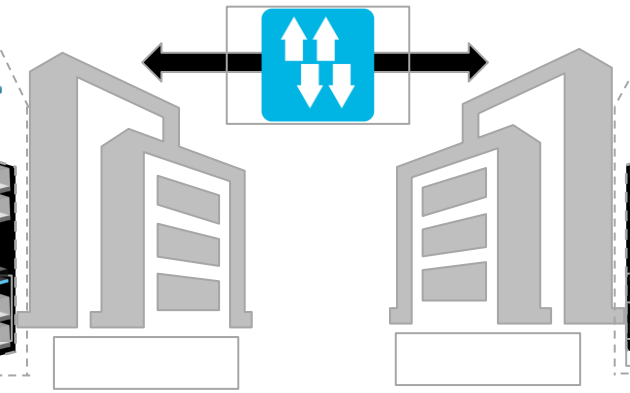
Scale-up

Across Racks



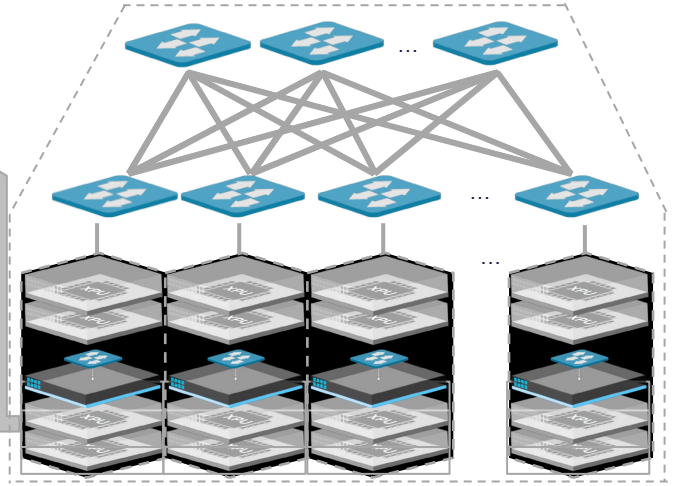
Scale-out

Across Data Centers



Scale-across

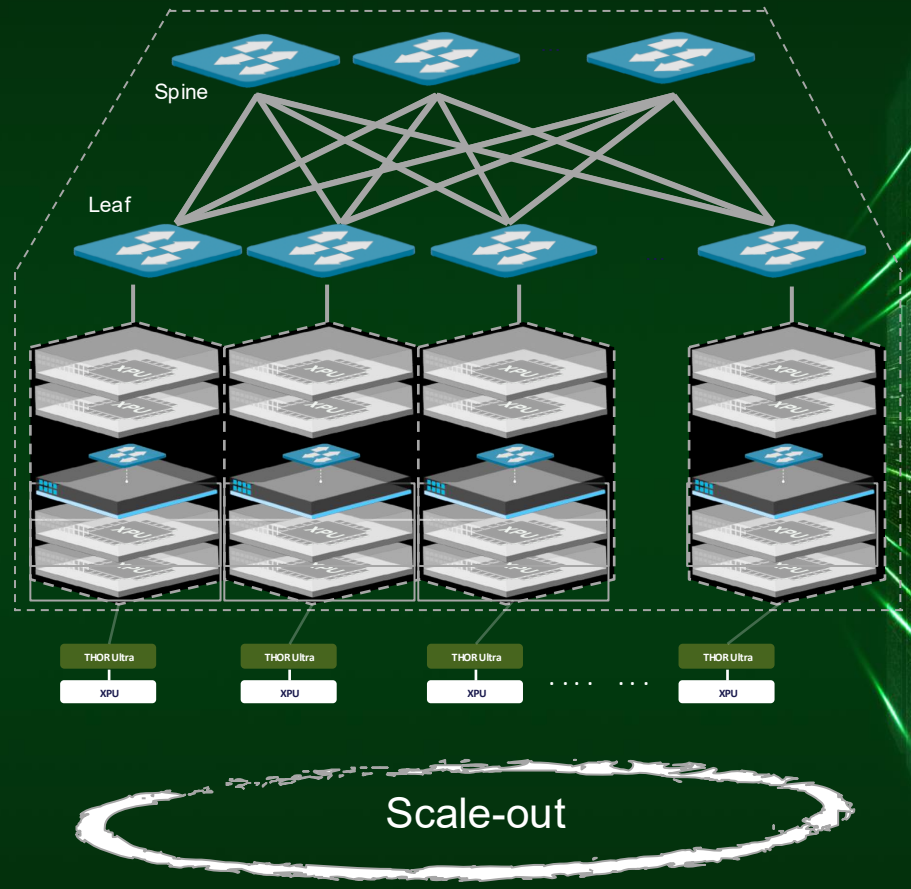
Across Racks



Scale-out

Scale-Out Backend Fabric – RDMA Traffic

- Elephant Flows
- Low traffic Entropy
- Bursty Traffic patterns
- Stragglers Impact overall performance



RDMA Not Built to Support AI @ Scale

- No multipathing
- No Out-of-order packet delivery
- Go-back-n → Inefficient
- DCQCN → Hard to tune
- Challenges with scale



MultiPath Reliable Connection – Enhanced RoCE

Packet Multi-pathing for efficient Load-balancing
Adaptive Packet Spray

Out-Of-Order Packet Delivery for maximum fabric utilization

Selective Retransmit for efficient data transfer

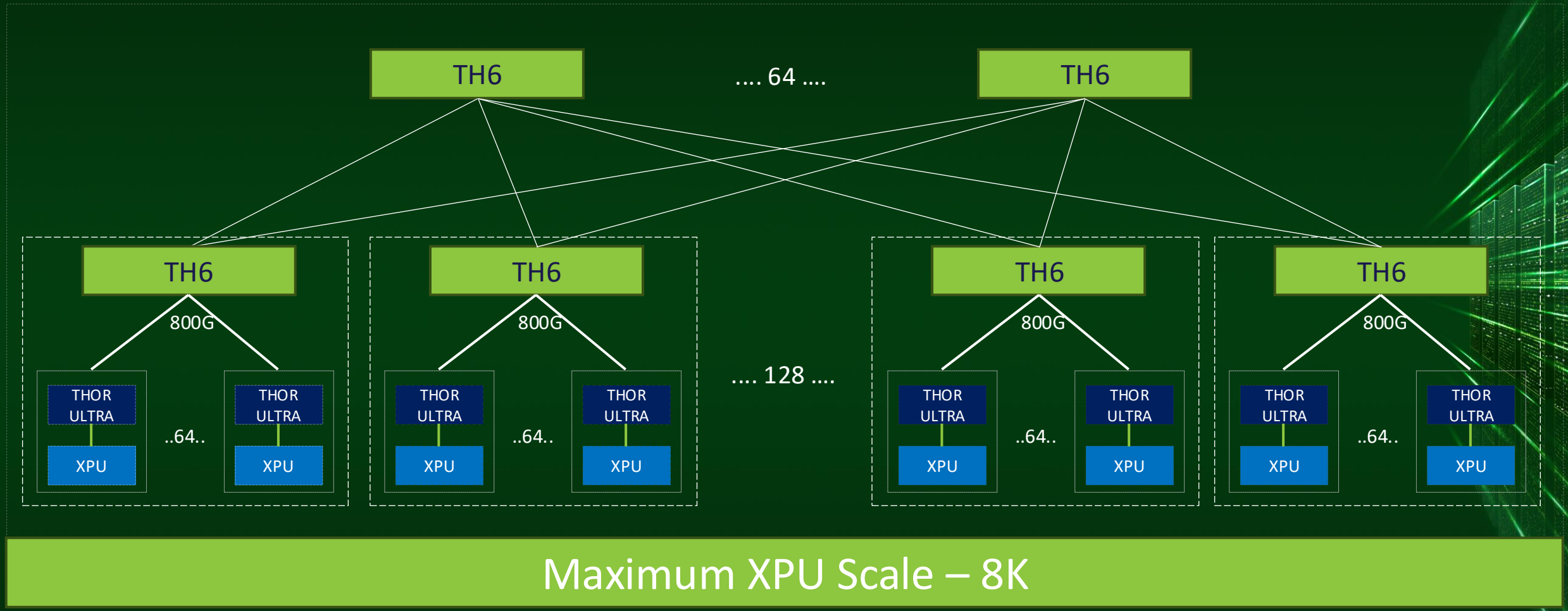
MODERNIZE RDMA

Programmable **Receiver & Sender based Congestion Control**
Minimize possibility of Congestion

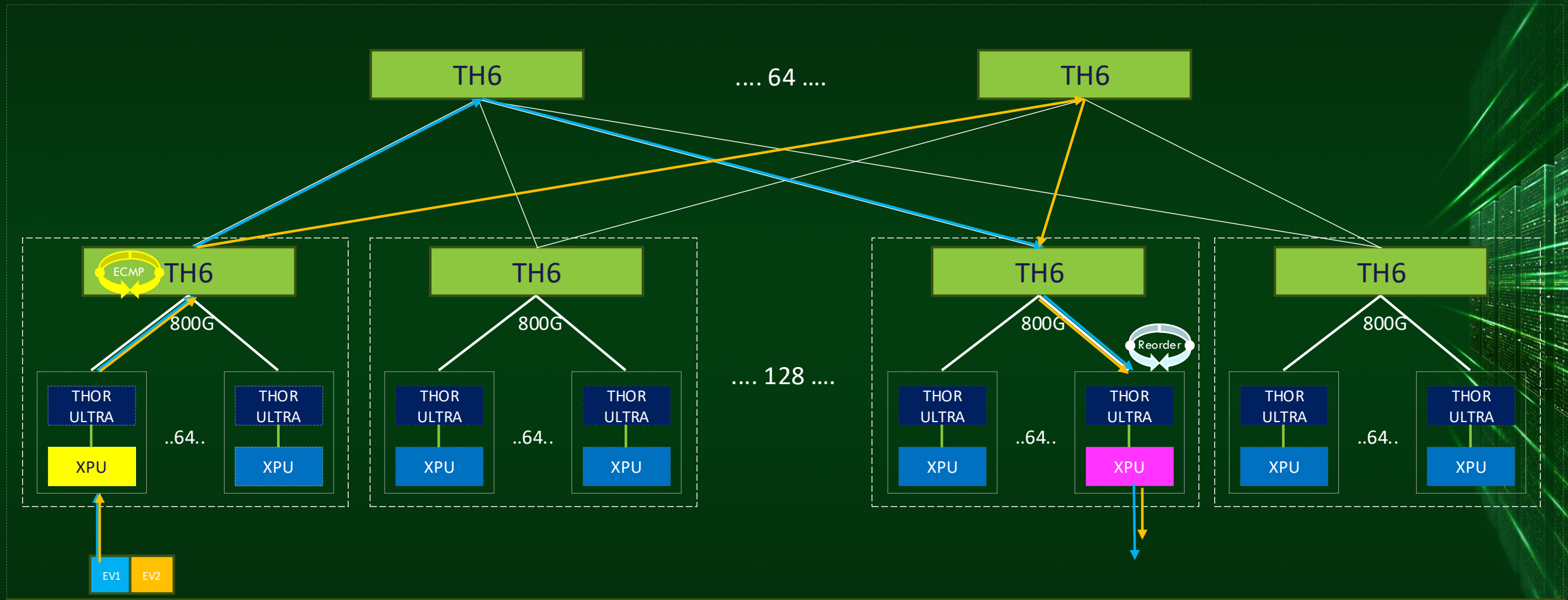
Advanced Telemetry across AI fabric
Reduce false positives with packet trimming

Source Routing with SRv6

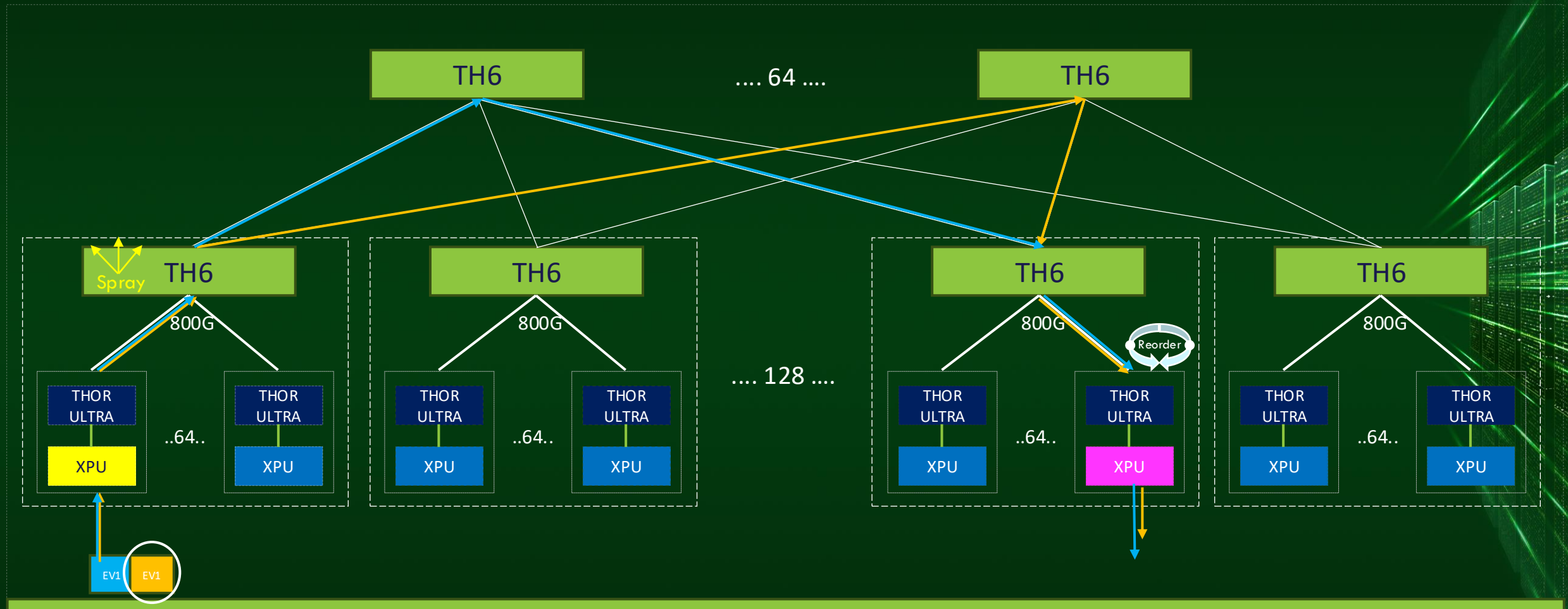
Scale-Out fabric - 2 Tier CLOS



2 Tier Scale-Out Fabric – Packet Flow



2 Tier Scale-Out Fabric – Packet Flow

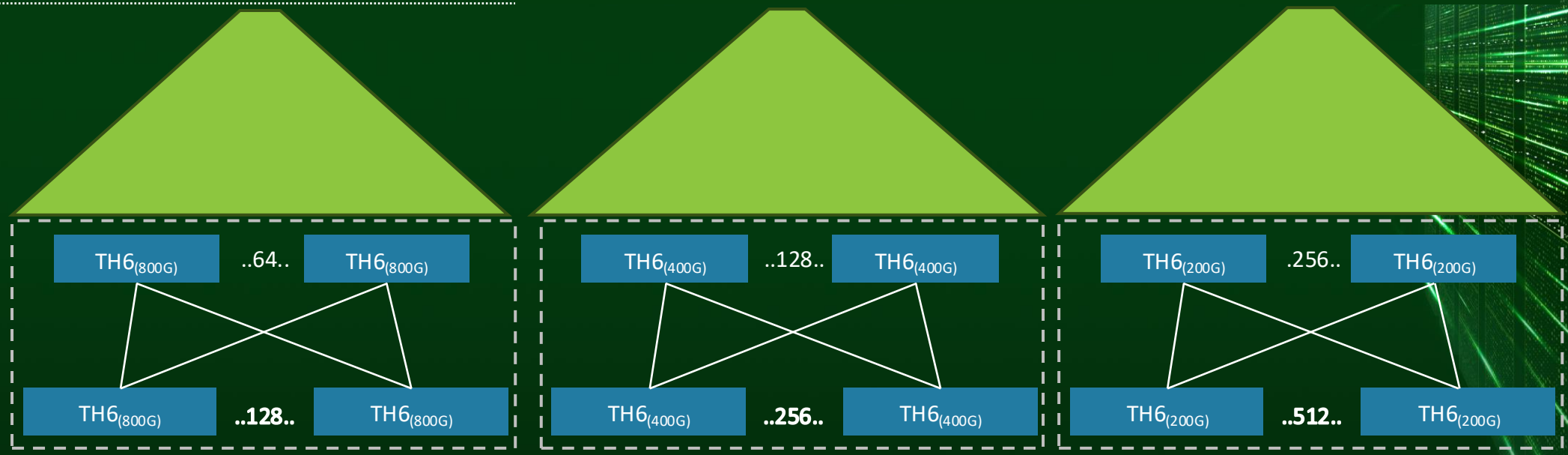


Multi-Plane Fabric Architecture



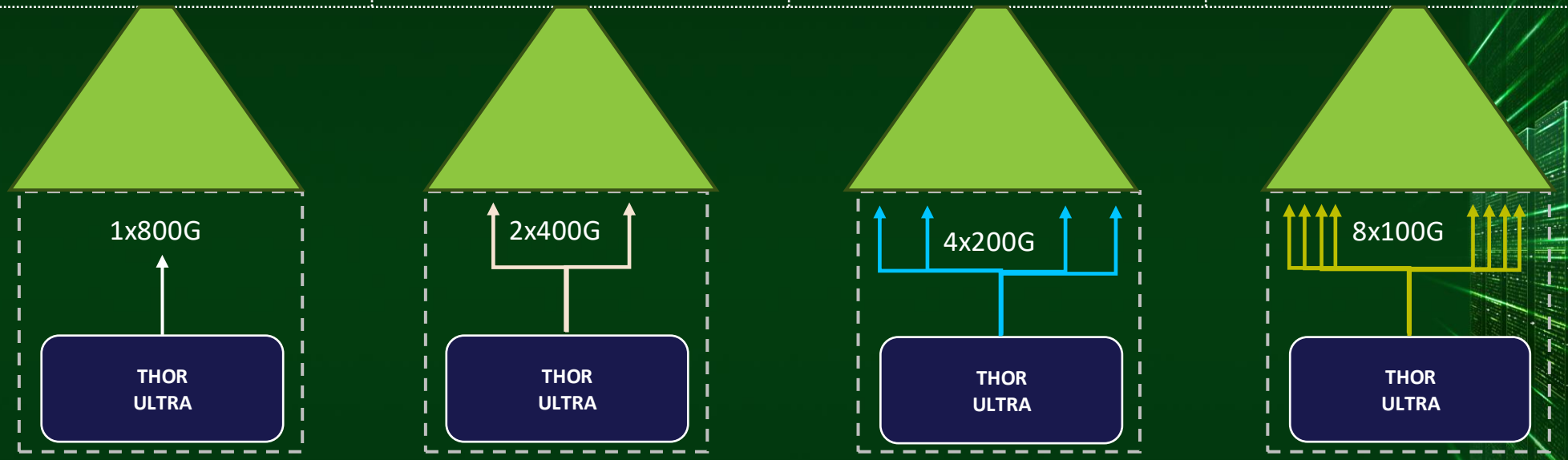
Switch – Port Break Out

Breakout (Port)	800G
TH6	128
CLOS Leaf's	128
CLOS Spines	64
Endpoints	8K



NIC – Port Break Out

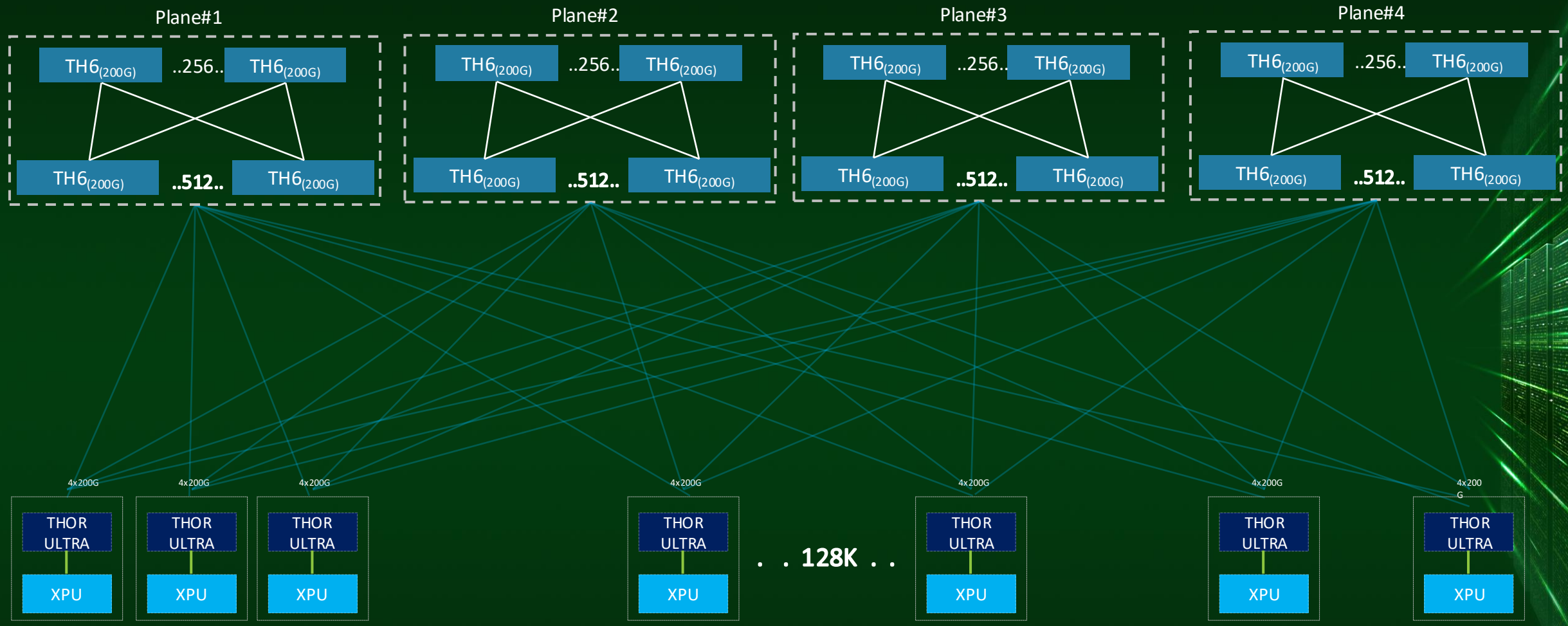
Breakout (Port)	800G	400G	200G	100
Thor Ultra	1	2	4	8



Multi-Plane Scale-Out Fabric

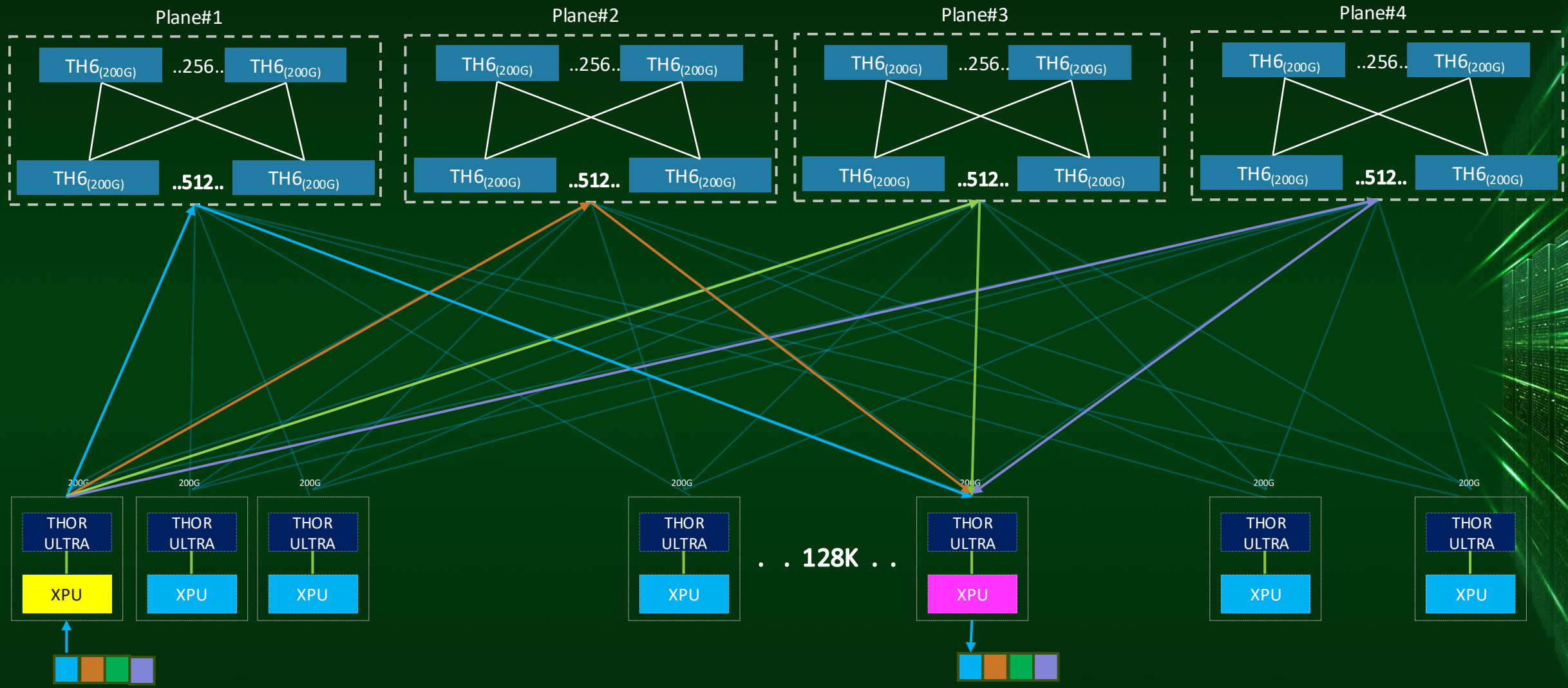


Multi-Plane Scale-Out Fabric



Maximum XPU Scale – 128K

Multi-Plane Scale-Out Fabric – Packet Flow



BGP based fabric routing or Source based routing Options

Design Considerations – Scale-Out Fabric

Packet Spraying	NIC Spraying	Switch Spraying
Fabric Planes	Single Plane	Multi-Plane
Routing	BGP based	Source based
Congestion Control	Sender based CC	Receiver based CC
Network Telemetry	Trimming, CSIG	ECN

Call To Action

- Learn about MRC at the OCP Academy
- Try MRC and share your experience
- Share your experience about your design choices for backend networks
- Share your make-RDMA-reliable@scale-wishlist with us
- Join the Enhanced RoCE & MRC effort @ UEC & OC


Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California

Scale-Up and Scale-Across Fabric

Scale-up

- ~10X Scale-Out bandwidth
- Low latency switches
- Optimized headers for memory Operations
- Lossless Ethernet / IP Fabric
  - CBFC
  - LLR

Scale-across

- Large distances ~kms (large RTT)
- Typically oversubscribed
- Large buffer switches